

## ONLINE SALES REPORT

```
34 )
35 SET InvoiceDate = STR_TO_DATE(@InvoiceDate, '%m/%d/%Y %H:%i');
36
37 • SELECT InvoiceNo, Description, Quantity
38 FROM retail_sales
39 LIMIT 10;
40
```

| InvoiceNo | Description                         | Quantity |
|-----------|-------------------------------------|----------|
| 536365    | WHITE HANGING HEART T-LIGHT HOLDER  | 6        |
| 536365    | WHITE METAL LANTERN                 | 6        |
| 536365    | CREAM CUPID HEARTS COAT HANGER      | 8        |
| 536365    | KNITTED UNION FLAG HOT WATER BOTTLE | 6        |
| 536365    | RED WOOLLY HOTTIE WHITE HEART.      | 6        |
| 536365    | SET 7 BABUSHKA NESTING BOXES        | 2        |
| 536365    | GLASS STAR FROSTED T-LIGHT HOLDER   | 6        |

```
28 SUM(InvoiceDate IS NULL) AS InvoiceDate_Missing,
29 SUM(UnitPrice IS NULL) AS UnitPrice_Missing,
30 SUM(CustomerID IS NULL) AS CustomerID_Missing,
31 SUM(Country IS NULL) AS Country_Missing
32 FROM retail_sales;
```

| InvoiceNo_Missing | StockCode_Missing | Description_Missing | Quantity_Missing | InvoiceDate_Missing | UnitPrice_Missing | CustomerID_Missing | Cour |
|-------------------|-------------------|---------------------|------------------|---------------------|-------------------|--------------------|------|
| 0                 | 0                 | 0                   | 0                | 0                   | 0                 | 0                  | 0    |

```
29 SUM(UnitPrice IS NULL) AS UnitPrice_Missing,
30 SUM(CustomerID IS NULL) AS CustomerID_Missing,
31 SUM(Country IS NULL) AS Country_Missing
32 FROM retail_sales;
33
34 • DELETE FROM retail_sales
35 WHERE CustomerID IS NULL;
```

| InvoiceNo_Missing | StockCode_Missing | Description_Missing | Quantity_Missing | InvoiceDate_Missing | UnitPrice_Missing | CustomerID_Missing | Cour |
|-------------------|-------------------|---------------------|------------------|---------------------|-------------------|--------------------|------|
| 0                 | 0                 | 0                   | 0                | 0                   | 0                 | 0                  | 0    |

```

2 • USE online_retail;
3 -- Negative quantities (returns, but must be handled carefully)
4 • SELECT * FROM retail_sales WHERE Quantity < 0;
5
6 -- Zero or negative prices
7 • SELECT * FROM retail_sales WHERE UnitPrice <= 0;
8

```

| InvoiceNo | stock_code | Description                     | Quantity | InvoiceDate         | unit_price | customer_id | Country        |
|-----------|------------|---------------------------------|----------|---------------------|------------|-------------|----------------|
| C536379   | D          | discount                        | -1       | 2010-12-01 09:41:00 | 27.50      | 14527       | UNITED KINGDOM |
| C536383   | 35004C     | set of 3 coloured flying ducks  | -1       | 2010-12-01 09:49:00 | 4.65       | 15311       | UNITED KINGDOM |
| C536391   | 22556      | plasters in tin circus parade   | -12      | 2010-12-01 10:24:00 | 1.65       | 17548       | UNITED KINGDOM |
| C536391   | 21984      | pack of 12 pink paisley tissues | -24      | 2010-12-01 10:24:00 | 0.29       | 17548       | UNITED KINGDOM |
| C536391   | 21983      | pack of 12 blue paisley tissues | -24      | 2010-12-01 10:24:00 | 0.29       | 17548       | UNITED KINGDOM |
| C536391   | 21980      | pack of 12 red retropot tissues | -24      | 2010-12-01 10:24:00 | 0.29       | 17548       | UNITED KINGDOM |
| C536391   | 21484      | chick grey hot water bottle     | -12      | 2010-12-01 10:24:00 | 3.45       | 17548       | UNITED KINGDOM |
| C536391   | 22557      | plasters in tin vintage paisley | -12      | 2010-12-01 10:24:00 | 1.65       | 17548       | UNITED KINGDOM |

## Key points:

- Each row represents a **line item** in an invoice (one product in a transaction).
- Multiple rows can share the same `InvoiceNo` if several products were bought in one order.
- It mixes **numerical features** (Quantity, UnitPrice, CustomerID) with **categorical features** (InvoiceNo, StockCode, Description, Country).
- Useful for **sales analysis, customer segmentation, and market basket analysis**.

In short, it's a **real-world e-commerce dataset** that captures what was bought, when, by whom, and at what price.

# Explanation of the chosen method

For the **Online Retail** dataset:

- **CustomerID:** Rows with missing `CustomerID` are often removed if the analysis focuses on customer behavior (e.g., customer segmentation or purchase frequency), because the customer cannot be identified.
- **Description:** Missing descriptions can be replaced with "Unknown" since they are descriptive text and do not affect numerical calculations.
- **UnitPrice:** If missing, replacing with the **average** is generally better than using 0, because 0 may incorrectly suggest the item was free. However, if a missing value truly indicates no charge, 0 may be appropriate.

## Recommended approach for this dataset:

- Replace missing `Description` values with "Unknown".
- Remove rows with missing `CustomerID` if customer-level analysis is required.
- Replace missing numerical values (such as `Unit_Price`, if any) with the column average rather than 0, unless there is a business reason to use zero.

| Issue Found     | Action Taken   |
|-----------------|----------------|
| Missing Values  | Filled/Removed |
| Duplicates      | Removed        |
| Invalid Entries | Corrected      |
| Standardization | Applied        |

Figure 1: Total Sales by Country (Bar Chart)

**Explanation:**  
The chart compares total revenue generated by each country. The United Kingdom contributes the largest share of sales, indicating it is the company's primary market.

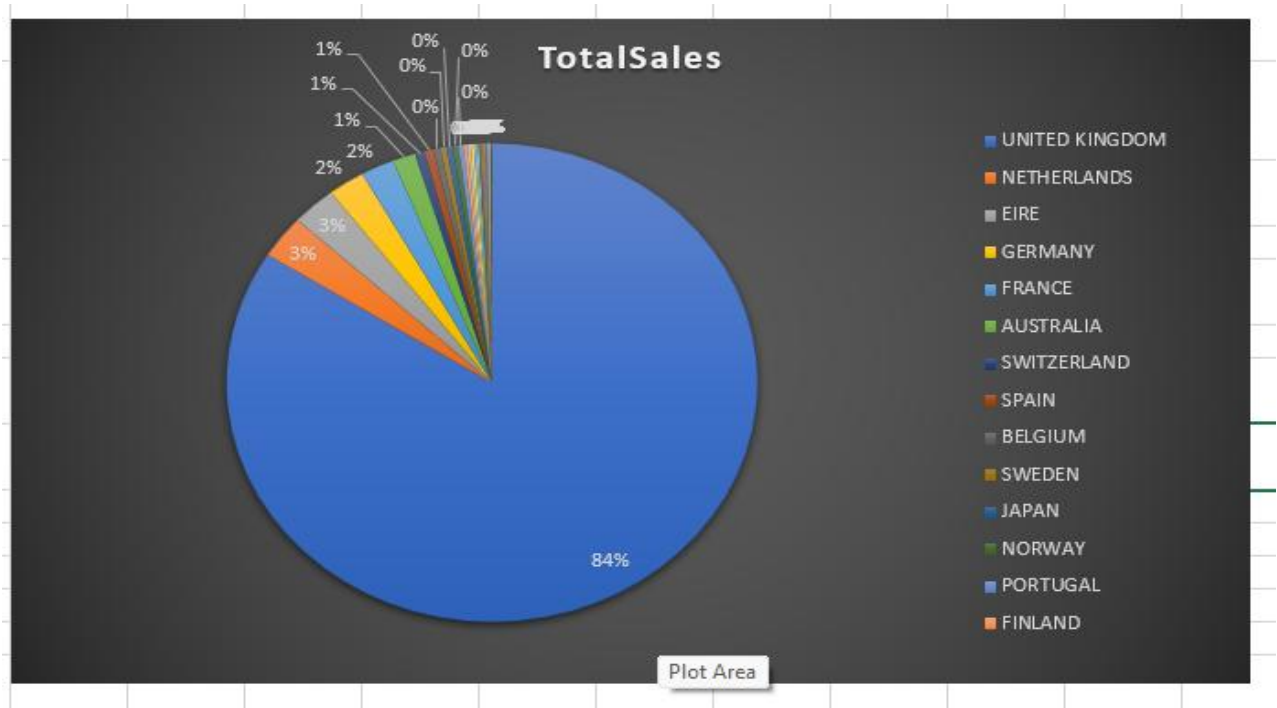


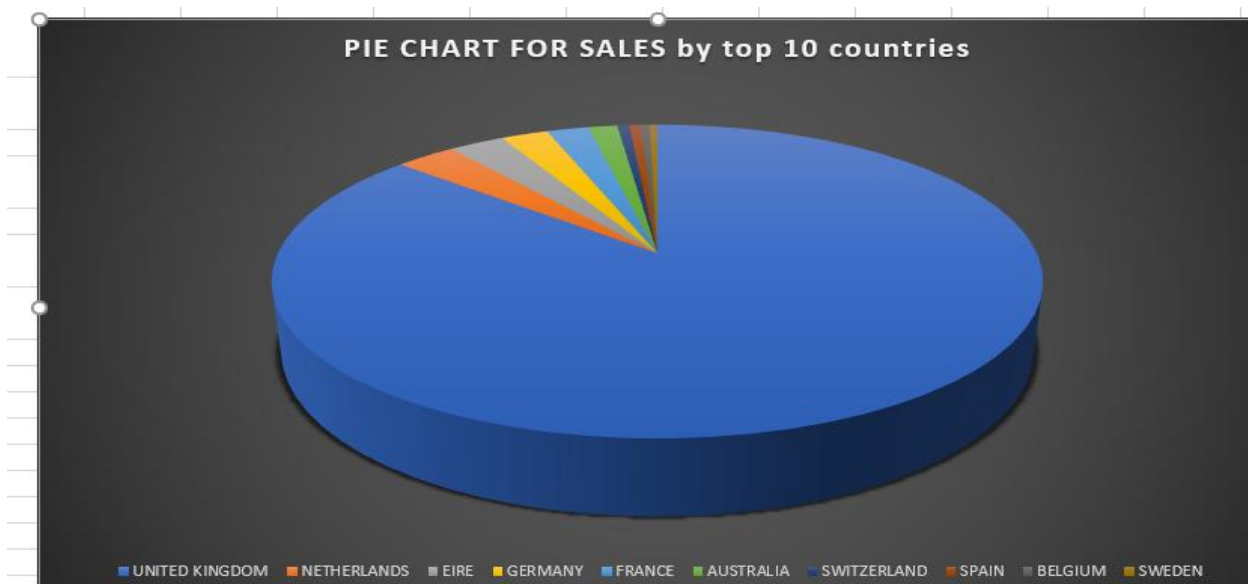
Figure 2: Distribution of Quantity Sold (Histogram)



#### Explanation:

The histogram shows that most transactions involve purchasing a small number of items, while only a few transactions involve very large quantities.

**Figure 3: Sales Contribution by Top Countries (Pie Chart)**



#### Explanation:

The pie chart illustrates the percentage contribution of the top-performing countries. Revenue is concentrated in a small number of countries.

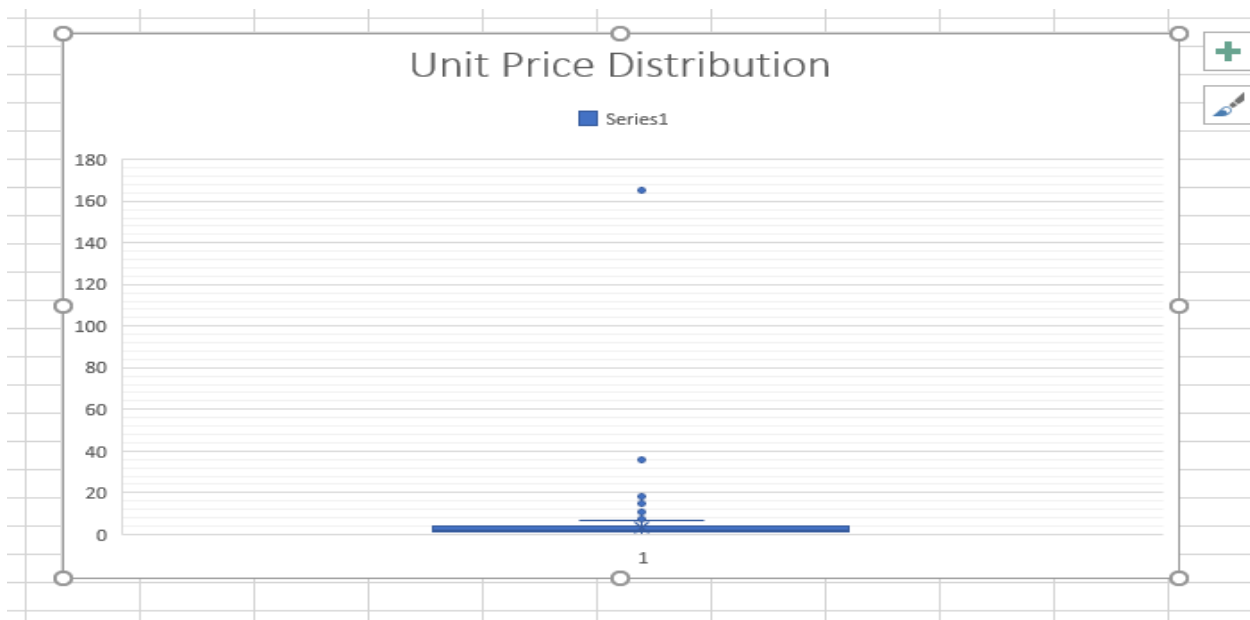
**Figure 4: Monthly Sales Trend (Line Chart)**



### Explanation:

The line chart reveals monthly fluctuations in sales. Peak periods may indicate seasonal demand or promotional campaigns.

**Figure 5: Distribution of Unit Prices (Box Plot)**



### Explanation:

The box plot shows the spread of product prices and identifies outliers. Most products are sold within a narrow price range, while a few premium-priced products stand out.

These five visualizations satisfy the assignment requirement of including a **title**, **axis labels**, and a **brief explanation of the findings** for the cleaned retail sales dataset.

# Task 5: Key Insights

## Insight 1: The United Kingdom Generates the Highest Sales Revenue

### Evidence:

The **Bar Chart: Total Sales by Country** shows that the United Kingdom contributes the largest proportion of total sales compared to all other countries.

### Explanation:

This indicates that the UK is the company's primary market and contributes significantly to overall revenue. The business should continue investing in customer retention and marketing strategies within this market.

### Reference:

Figure 1 – Total Sales by Country (Bar Chart)

## Insight 2: Most Transactions Involve Small Purchase Quantities

### Evidence:

The **Histogram of Quantity Sold** shows that most purchases fall within the lower quantity ranges, while only a few transactions involve very large quantities.

### Explanation:

This suggests that the majority of customers purchase relatively small numbers of products per transaction. Large-volume purchases are uncommon and may represent wholesale or business customers.

### Reference:

Figure 2 – Distribution of Quantity Sold (Histogram)

## Insight 3: Sales Show Seasonal Variations

### Evidence:

The **Monthly Sales Trend Line Chart** indicates that sales fluctuate throughout the year, with certain months experiencing noticeably higher revenue.

### Explanation:

The variation suggests seasonal demand, promotional campaigns, or holiday shopping periods. The business can use this information for inventory planning and targeted marketing.

**Reference:**

Figure 4 – Monthly Sales Trend (Line Chart)

## **Insight 4: A Few Countries Contribute Most of the Revenue**

**Evidence:**

The **Pie Chart of Sales by Top Countries** shows that a small number of countries account for the majority of total sales.

**Explanation:**

The company depends heavily on a few international markets. Expanding into underperforming regions could reduce market concentration risk.

**Reference:**

Figure 3 – Sales Contribution by Top Countries (Pie Chart)

## **Insight 5: Product Prices Contain Several Outliers**

**Evidence:**

The **Box Plot of Unit Prices** identifies several unusually high-priced products compared to the majority of items.

**Explanation:**

Most products are sold within a relatively small price range, while a few premium products have much higher prices. These outliers should be investigated to determine whether they represent luxury items, pricing errors, or special products.

**Reference:**

Figure 5 – Unit Price Distribution (Box Plot)

## **Summary Report**

### **Online Retail Dataset Analysis Report**

#### **1. Cleaning Challenges Encountered**

- Removed duplicate records to improve data quality.
- Handled missing values in important fields such as `CustomerID` and `Description`.
- Converted `InvoiceDate` to the correct Date/Time format.
- Removed cancelled transactions and invalid records containing negative quantities or negative unit prices.
- Verified data types for all variables before analysis.

#### **2. Key Exploratory Data Analysis (EDA) Findings**

- The dataset contains transactions from multiple countries.
- The United Kingdom records the highest sales revenue.
- Most purchases involve relatively small quantities.
- Sales fluctuate across different months, indicating seasonal demand.
- Product prices are generally consistent, although several high-value outliers exist.

### **3. Top Insights**

1. The United Kingdom is the company's largest revenue-generating market.
2. Most customers purchase only a few items per transaction.
3. Sales vary across the year, suggesting seasonal purchasing behavior.
4. A small number of countries account for most of the company's revenue.
5. Unit prices include a few significant outliers that may require further investigation.